IOT PROJECT · GROUP 13 · TECHNION

MindBox

A smart focus box that **measures your study environment in real time** — and helps you actually focus.

What it is: an ESP32-S3 device + companion web app for focused studying.

Board: LCDWIKI ES3C28P (ESP32-S3, 2.8" touchscreen) · **Firmware:** 0.2.0-s3

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A short, plain-English guide: what MindBox does, how it works, how it survives things going wrong, and how to see it for yourself.

Contents

> **How to read this:** Blue boxes explain things in plain English · amber boxes hold the one key idea · green boxes show how to see a feature live · red boxes are watch-outs.

1 The big picture

MindBox is a box you keep on your desk. It runs your study timer, senses your environment, and coaches your focus — working with a website that shows your history and controls the box from anywhere.

> **In one line:** A desk device that **measures** your study conditions (noise, light, temperature, and whether you're there), runs focus sessions, and syncs to a website — even when the Wi-Fi drops.

> **WHY IT EXISTS**
> Most study timers are passive — they count minutes but don't know if you're really focusing. MindBox turns your environment into something you can **see and measure**: the difference between a stopwatch and a coach who notices the room got loud and you wandered off.

1.1 What the box does

⏱ Runs sessions
Pomodoro work/break timers you set with a physical knob — no phone needed.

🎤 Senses
Noise, temperature, humidity, light, and whether you're at the desk.

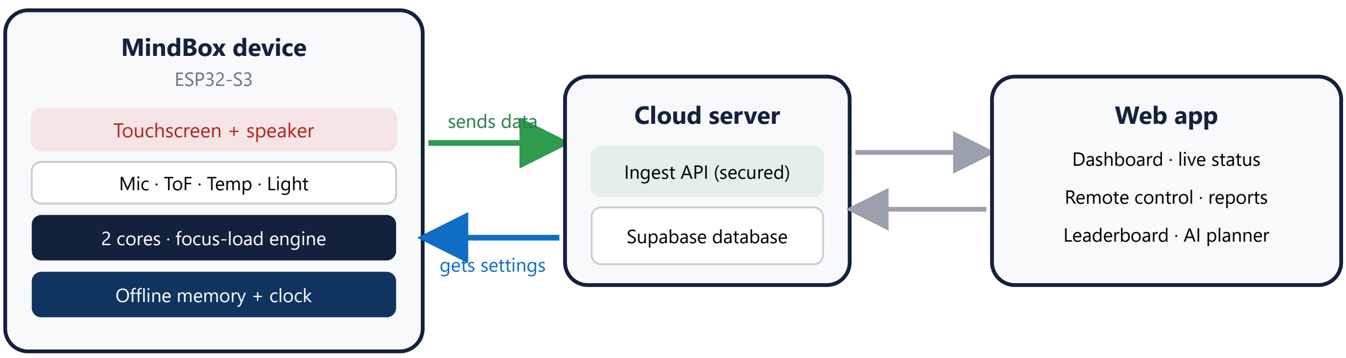
📊 Scores focus
Blends those signals into a live 0–100 "focus load" number.

📡 Syncs
Sends sessions to a website for history, stats, streaks, remote control.

🔌 Works offline
Keeps running with no internet, then syncs automatically when it's back.

🤝 Shares
Friend leaderboards, and read-only reports for a parent, tutor, or coach.

1.2 How the pieces fit together



The **device** senses and runs sessions · the **server** stores everything · the **web app** shows it and controls the box.

1.3 A study session, start to finish

1. **Set up** — turn the knob to pick a duration; tap to start (no phone).

2. **Focus** — the box senses noise, light, temperature, and your presence while a hardware timer counts down.
3. **Coach** — room gets loud or you leave? It reacts: a chime, an "away" pause, a focus-load bump.
4. **Finish & sync** — a completion chime and on-screen summary; the session (with minute-by-minute data) uploads to the website. No Wi-Fi? It's saved and uploads later.

> **SEE IT LIVE**
> Open the `/device` page next to the box — the online dot, battery, and "Room right now" readings update within seconds of the box sending data. The clearest proof the whole chain works.

2 How it's built

A dual-core ESP32-S3 touchscreen board with a handful of everyday sensors — plus a few smart software decisions that keep it fast and reliable.

> **In one line:** Sensors on a 4-pin-only board (so two of them **share a pin**), driven by **two processor cores** — one keeps the screen smooth, the other handles the slow internet in the background.

2.1 The sensors

Part	What it's for	Talks over
ILI9341 touchscreen	timer, status, menus (tap & swipe)	SPI
FT6336 touch	reads your finger	I2C
VL53L1X ToF	are you at the desk? (laser distance)	I2C
ES8311 mic	room noise level	I2S
Speaker + amp	chimes & "ring my device"	I2S
KY-040 encoder	set duration (turn) + button (press)	digital
DHT11	temperature & humidity	1-wire
KY-018 light	brightness → auto-dims the screen	analog

On: touch presence mic speaker light temp Wi-Fi **Replaced by the screen + speaker:** LED ring vibration physical toggle

2.2 The clever bit: two sensors on one pin

The board breaks out only **four** spare pins, so **IO21 does double duty**: it's both the encoder's push-button *and* the DHT11 temperature sensor's data line.

> **KEY IDEA**
> The button and the temperature sensor **share one pin** — safe because the DHT11's pull-up holds the line "high" (just what the button needs), and the firmware simply **skips a temperature read while the button is pressed**. The knob's *rotation* is on different pins, so it never clashes. `config.h:117, config.h:122`

2.3 Two cores keep it smooth

> **IN PLAIN ENGLISH**
> One worker (**Core 1**) keeps the screen, touch, and sensors responsive. A second worker (**Core 0**) handles slow Wi-Fi and audio in the background. A separate **hardware timer** counts the session down — so even a slow 18-second internet call never freezes the screen or loses a second.

2.4 Why the screen never freezes

Only Core 1 draws — into one off-screen image that's pushed to the display in a single fast copy. Because there's a single writer, there's **no tearing and no need for locks**. The image is kept small (8-bit colour) to leave memory free for Wi-Fi.

2.5 The session is a state machine

The box moves through clear states — the human flow is:

```
BOOTING → IDLE ⇄ RUNNING ⇄ PAUSED → COMPLETE → (HOMEWORK) → IDLE
                   ↑                                 ↓
                RESUME  ←— (power cut, restored)    ERROR (hardware fault)
```

SEE IT LIVE

Turn the knob (menu moves), press it (selects), breathe on the sensor (temperature rises) — three things through one shared pin, no conflict. And drop the Wi-Fi mid-session: the screen and timer keep going without a stutter (that's Core 0 taking the hit, not Core 1).

3 The Focus-Load number

Behind all the sensors sits one small, honest formula that folds everything into a single 0–100 score you can read at a glance.

> **In one line:** The **Focus Load Estimate (FLE)** mixes how far you are into the session, how noisy the room is, how much the light flickers, and whether you're at the desk — an **estimate**, not a brain reading.

> **IN PLAIN ENGLISH**
> It's like a "conditions" score — the way a weather app turns temperature, wind, and humidity into one "feels like" number. **Low** = calm, easy conditions. **High** = the environment is working against you.

3.1 The formula

```
FLE = round( 100 × ( 0.30·progress + 0.30·noise + 0.20·lightVariance + 0.20·presence ) )
```

Each input is squeezed to 0–1 first. The weights are the four ingredients' share of the 100 points:

Progress · 30
How far into the session (elapsed ÷ target) — a gentle fatigue proxy.

Noise · 30
Room loudness from the mic (0 = library, 1 = loud).

Light variance · 20
How much the light flickers (steady = calm, jumpy = higher).

Presence · 20
Are you at the desk (count capped at 5, then scaled).

Source of truth: `focus_load.h` — the same formula runs on the device, the simulator, and the website (drift-guarded), so the number is identical everywhere.

3.2 What the numbers feel like

Situation	FLE
Fresh, quiet, at the desk	**20**
Mid-session, moderate noise, present	**52**
…then you leave the desk	**32** (−20)
Late, loud, flickering light, present	**90**

Rules of thumb: a loud room adds up to **+30**; leaving the desk removes **−20**; every 0.1 of extra noise ≈ **+3**.

ADAPTIVE COACHING

When the score climbs to **75**, the box steers you toward a break (a shorter work interval), rate-limited to once every 5 minutes so it never nags.

SEE IT LIVE

Watch the number on the screen (or Serial `m`): **clap** and it jumps (noise ↑); **walk away** and it drops ~20 (presence ↓) — matching the rules above.

4 Cloud, app & the 16 features

The box talks to a server over a few small, secured messages; the website turns that into live dashboards, remote control, and everything the 16 user stories promise.

> **In one line:** The box is like a phone texting short, secret-signed updates to a server; the website reads them and can text back settings and commands.

4.1 How the box talks to the cloud

Every message carries a shared secret header — a wrong or missing secret is rejected (**401**).

Message	Direction	What it carries
sessions	box → server	a finished session (+ minute-by-minute data)
telemetry	box → server	a 30-second heartbeat: state, battery, Wi-Fi, room conditions
config	server → box	every 60s: the box's settings + any remote command
pairing / unpair	box → server	claiming / releasing the box
homework	box → server	homework progress bumps

HONEST DETAIL

The website's "live" updates are **polling** (it asks the server every few seconds), not push — simpler, and easy to demonstrate.

4.2 What you see on the website

Screen	Shows
`/device`	online dot, battery, Wi-Fi, "Room right now" — updates ~every 10s
Dashboard	today's focus, streak, device state
Session detail	minute-by-minute focus + environment graphs
Remote control	Start / End / Ring / Message (Queued → Sent → Done)
Leaderboard · Exports · Reviewers	friend streaks · PDF/CSV/email reports · read-only sharing
Calendar	schedule + a **"Smart (AI)" planner** (Gemini)

4.3 The 16 user stories at a glance

#	Story	What it does	
1	Physical Controls	knob + button + touch to run sessions	done

2	Hardware Feedback	screen + chimes (replaces LED ring + buzzer)	evolved
3	Local Timer Setup	set duration with the knob, offline	done
4	Offline Buffering	save sessions with no Wi-Fi, sync later	done
5	Session History	list, graphs, date range, focus streak	done
6	Multi-User Leaderboard	streak ranking among friends	done
7	User Identification	switch profiles on the box	done
8	Environmental Sensing	log noise, temperature, light	done
9	Presence Detection	pause/alert when you leave the desk	done
10	Focus Load Estimation	the live 0–100 score (Chapter 3)	done
11	Export Delivery	download/email PDF & CSV reports	done
12	Motivation Nudges	a nudge after long inactivity	done
13	Read-Only Reviewer	share reports with a parent/tutor	done
14	Wi-Fi Setup	connect via on-screen setup	done
15	Device Status Signals	boot/status on the screen (replaces LED ring)	evolved
16	Adaptive Interval Coaching	adjust work/rest by focus load	done

HARDWARE EVOLUTION

The original plan's LED ring, vibration motor, and physical toggle became the **touchscreen** (visual status) and the **speaker** (chimes) + touch mode-switching. Same jobs, richer hardware.

5 Edge cases & resilience

Real desks are messy: the Wi-Fi drops, someone trips over the USB cable, a sensor comes loose. MindBox is built to survive all of that — and to **show you** that it did.

In one line: Nothing is lost when things go wrong: the session keeps running, the data is saved locally, and it all syncs the moment things recover — with a visible trail so you can prove it.

5.1 What happens if the internet drops?

IN PLAIN ENGLISH

The box doesn't need the internet to do its job. If the Wi-Fi falls mid-session, **the timer keeps counting** (it runs on a hardware clock, not on the network), the session data is **saved to the box's own flash memory**, and the box quietly retries the connection every 15 seconds. When Wi-Fi returns, the saved session **uploads automatically** — nothing is lost.

During the outage

Session runs normally on screen. Finished sessions queue up in on-device storage. The box retries Wi-Fi every 15s.

When it comes back

Queued sessions upload oldest-first. If the clock hadn't synced yet, the times are corrected from internet time (NTP).

No duplicates

Each session has a unique id, so re-sending after a flaky connection never creates a double.

WHAT TO LOOK FOR

Kill the hotspot mid-session → the screen & timer keep going. Type Serial `q` → the session shows as **queued**. Turn Wi-Fi back on → within a minute it appears on the website with the correct times. The `/device` online dot goes grey after ~90s of silence, then green again on reconnect.

5.2 The full resilience table

When this goes wrong...	What the box does	What to look for
Wi-Fi / internet drops	Timer keeps running (hardware clock); sessions buffer to flash; retries every 15s; auto-syncs on return.	Screen keeps counting; Serial `q` = queued; appears online after reconnect.
Power cut mid-session	A checkpoint is saved to non-volatile memory on every start/pause/resume.	On reboot: the reset reason + a **"Resume session?"** prompt.
A sensor fails / unplugs	Keeps the last good value; clamps out impossible readings; a real fault flips to an ERROR state.	Red **Fault** row on `/device`; the status badge turns "danger".
Sensor bus locks up (I2C)	The presence sensor is a "canary" — pinged every 2s; if the bus jams, the box auto-recovers it (pulses the clock line); if it can't, it shows ERROR.	Recovers silently; ERROR screen only if truly stuck.

Weak USB supply (brownout)	The Wi-Fi power spike can sag a weak 5V rail. We cap the radio's power to soften it; the honest real fix is a stronger supply.	Boot screen shows reset reason **"brownout"** — so a power sag is never mistaken for a code crash.
Clock not set yet (offline boot)	Offline sessions are stamped with a placeholder time, then corrected from internet time once online.	Times look right on the website after the first sync.
Unauthorized data / wrong secret	Every upload must carry the shared secret header; anything else is refused.	A request without the secret returns **401**; with it, the real data.
Slow / half-open server call	Network work runs on the second core with a watchdog; a stalled call can't freeze the UI or the timer.	Screen stays smooth even while the box struggles to reach the server.

THE DESIGN PRINCIPLE

Every failure has a **graceful fallback** and a **visible signal**. The box never silently loses data, and it always tells you (on screen or on the site) what went wrong — which is exactly what you want to be able to demonstrate.

6 See it work

Every claim in this document can be checked in seconds. Here's what to do and what you'll see — plus ready-made accounts so the website is full of data on day one.

> **In one line:** Open `/device` next to the box, run a short session, and watch the readings, graphs, and remote controls respond live.

6.1 Claim → proof

Claim	Do this	You'll see
Reports to the cloud every 30s	open `/device`, wait	green Online dot; "Room right now" updates
Real mic measures noise	clap near the box	Serial `a` RMS jumps; value rises on the site
Presence sensor works	wave a hand; walk away in a session	distance changes; timer shows "Away"
Temp / humidity	breathe on the DHT11	temperature climbs within seconds
Light + auto-dim	cover the light sensor	reading drops; the screen dims
Site controls the box	"Ring my device" on `/device`	box chimes; status Queued → Sent → Done
Offline + sync	kill Wi-Fi, run a session, reconnect	queued, then appears with correct times
Survives power loss	yank USB mid-session, replug	boots to "Resume session?"
Data is authenticated	call the API without the secret	rejected with **401**

6.2 Demo accounts

Pre-loaded on the live website so every history/stats/leaderboard feature has data. **Password for all:** `MindboxDemo2026!`

Email	Use it to show
`mindbox.primary@example.com`	**Main account** — 30-day history, streak, focus & environment graphs, homework, courses, schedule. **Pair the box to this one.**
`mindbox.friend1@example.com`	a friend with a streak → the leaderboard
`mindbox.friend2@example.com`	a second friend → the leaderboard
`mindbox.reviewer@example.com`	read-only reviewer view of the main account
`mindbox.new@example.com`	an empty account → onboarding + a motivation nudge

> **BE HONEST IN THE DEMO**
> The demo-account data is generated by the project's own **simulator** (itself a documented feature) — fine to present, just not real personal history.

THE 30-SECOND PROOF

Log into the main account, open `/device`, and start a short session on the box. Watch the live readings move, the session graph fill in, and a "Ring" command light up the box — the whole device → cloud → website loop, in front of you.